Mahmoud El Walid Abdelkarim

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EDUCATION

Grand Valley State University · Allendale, MI — Expected Graduation: 04/28

B.S. Artificial Intelligence, Minor in Business · GPA 3.8 · Honors College

- Coursework: Data Structures and Algorithms, Discrete Mathematics, Data Visualization, Applied Statistics.
- NCAA Division II Swim & Dive: **2026 NCAA All-American** (7th, 100 back) and **2026 GLIAC Champion** (100 back), plus GLIAC relay titles in the 200 medley and 400 free.

EXPERIENCE

Rowad Modern Engineering (RME) · Cairo, Egypt — Jun – Jul 2026

AI Integration Intern, Strategy & PMO

- Led **RME Intelligence** from discovery to launch: an internal AI platform with **13 modules**, now used by **38 people across 5 construction projects**. Ran stakeholder interviews with Flex Labs and sequenced the build by business impact.
- **Data pipeline:** bilingual (Arabic/English) payment-certificate PDFs → Claude structured-JSON extraction → validation engine → auto-filled Oracle Unifier invoice template. Replaced manual re-keying with a **15–40 second** read per certificate.
- **Data pipeline:** built a BOQ revision-diff engine that matches ~450 line items by work package, so quantity variances come from arithmetic, not the model. Added RAG search: chunk → embed → pgvector → answers with page citations.
- **Web app:** Next.js and TypeScript on Vercel and Supabase, with 25 screens, 5 API routes, 4 access roles and an executive dashboard for contract vs. certified value, claims exposure and risk flags.
- Wrote the platform audit: scored 9 dimensions, logged **14 risks** (auth, persistence, serverless timeouts) and set the fix order. Drafted **10 KPIs** for the Tech Strategy team to measure completed initiatives.

Emerge Capital · Dubai (Remote) — Exposure Program

Venture Builder

- Supported market research, pitch content and product ideation for AI and Web3 portfolio ventures.
- **FINU** (meetfinu.com): AI money assistant that tracks spending, flags bad habits and helps users set and hit goals, with Autopilot savings and the Finu Card. Backed by **$200K in Google Gemini credits**.
- **Web app:** built FINU's marketing and demo site in **6 days** (27 commits, ~8,700 lines): Next.js 15, React 19, TypeScript and a dependency-free WebGL shader that pauses off-screen, caps pixel density on mobile and falls back without WebGL.

PROJECTS (FOUNDER, 2024 – PRESENT)

FORGE · AI training app for athletes — Sep 2026 – Present

- Shipped a **34,000-line** TypeScript app in 6 weeks (46 commits), run from 13 written architecture specs.
- **Data pipeline:** workout photo → background job → Claude tool call checked against a Zod schema → one retry → confidence flags → athlete confirms. Reads take 13–31 s; idempotent request IDs make retries free.
- **303 automated tests** (Vitest, Playwright), a labeled-photo eval set, 18 SQL migrations with row-level security, WHOOP sync.

NYX · nyxsystem.online · multi-agent AI system — Jun 2026 – Present

- **Agent architecture:** one Claude Opus orchestrator directs 5 Claude Sonnet specialist agents through a streaming workflow engine that chains each step's output. 54 API routes, scheduled briefings, Google/Notion integrations.
- Audited the platform and raised its health score from **71 to 100**: closed 2 security holes, fixed a bug that dropped every waitlist signup, and added rate limiting plus a CI pipeline.

TONED · operating platform for Egypt's first hot pilates studio — Jun – Sep 2026

- Built a React Native member app, staff admin panel and Supabase backend with card and wallet payments and an AI concierge that answers from live schedules. Wrote the import for **1,469 legacy client records**.

SKILLS

Agentic engineering: Claude Code and OpenAI Codex driven by per-repo specs (CLAUDE.md, AGENTS.md), MCP servers, multi-agent orchestration, tool calling, structured outputs, RAG, evals
Languages: Python, TypeScript, JavaScript, SQL · Arabic (native), English (fluent)
Stack: Next.js, React, React Native (Expo), Supabase (Postgres, pgvector, Edge Functions), Vercel, Netlify, GitHub Actions, Vitest, Playwright
Program management: Requirements and stakeholder interviews, risk registers, phased roadmaps, KPI design, executive reporting